Nanyang Technological University
School of Electrical & Electronic Engineering
IE2006 Engineering Mathematics I

Tutorial 7 Probability & Statistics

1. A diet program states that the participants are expected to lose over 22 pounds in 5 weeks. Suppose it is known that the standard deviation of weight loss (in 5 weeks) of participants of this program is 10.2 pounds. 20 participants are selected at random and the mean of weight loss (in 5 weeks) is found to be 23.5 pounds. Could this finding substantiate the claim of the diet program? Test with $\alpha = 0.05$. State any assumptions needed.

2. A new brand of canned mixed nuts plans to enter the Singapore market. The label says that 25% of the contents are cashews. Suspecting that this might be an overstatement, an inspector takes a random sample of 31 cans and measures the percentage weight of cashews [i.e., 100(weight of cashews)/(weight of all nuts)] in each can. The mean and standard deviation of these measurements are found to be 23.5 and 3.1, respectively. Do these results constitute strong evidence in support of the inspector's belief? Test with $\alpha = 0.01$.

3. The manager of a production facility claimed that he did not meet the production orders last month because the variance in production times of different workers was too great, resulting in too many delays. He said the standard deviation was over 30 min for the workers in his plant. You must decide if he is right, so you sample the production times of 25 workers from his plant and find that $s = 23$ min. What do you think of his claim? Use $\alpha = 0.05$. State any assumptions needed.

Answers

1. Claim is not substantiated
2. Inspector's belief is supported
3. Claim is not substantiated